YUROU CELINE XIAO

Ph.D. in Mechanical and Industrial Engineering
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Education

Doctor of Philosophy in Mechanical and Industrial Engineering September 2020 – October 2025
SintonLab, University of Toronto Toronto, ON

- Fast-Track Transfer from the MASc program
- Thesis: *Electrochemical Systems for Reactive Capture of CO_2*. Supervised by Prof. David Sinton
- Cumulative GPA: 4.0/4.0

Bachelor of Science in Mechanical Engineering September 2015 – April 2020
University of Alberta Edmonton, AB

- Cooperative Program with Distinction
- Cumulative GPA: 3.9/4.0

Research Experience

Doctoral Researcher September 2021 – October 2025
SintonLab, University of Toronto Toronto, ON

- Designed and synthesized selective, stable electrocatalysts for the integrated capture and electrochemical conversion of CO_2.
- Developed energy-efficient systems with temperature and pressure controls for integrated electrochemical systems.
- Research program leader, sponsored by Shell International B.V. and the Government of Canada, for the research project entitled "Electrosynthesis of Carbon Monoxide from Capture Fluids" (May 2023 – August 2025).
- Capture sub-group leader, managed a team of up to 12 researchers across research themes of direct air capture, flue gas utilization, and integrated CO_2 capture and conversion (May 2023 – August 2025).
- Electrolyzer fabrication coordinator, served as the primary technical liaison between researchers and machinists (September 2023 – August 2025).
- Co-founder of E-quester, a startup team that developed a low-cost and scalable electrically powered direct air capture system for CO_2 removal, resulting in a $250,000 Student Prize and named a Top 60 Milestone Award Finalist in the XPRIZE Carbon Removal competition (September 2021 – October 2022).
- Mentored 2 graduate students, resulting in co-first authored publications.

Doctoral Researcher January – April 2024
CSIRO Energy Center Newcastle, Australia

- Developed and scaled CO_2 electrolyzers from 1 cm^2 to 100 cm^2 for integrated CO_2 capture and conversion systems.

Graduate Researcher September 2020 – August 2021
SintonLab, University of Toronto Toronto, ON

- Pilot-plant operator for CERT, a startup team that scaled a CO_2-to-ethylene electrolyzer system to 40,000 cm^2 of total active area as a Top 10 Finalist in the NRG COSIA Carbon XPRIZE Competition.

Energy Analysis Intern May 2020 – March 2021
University of Calgary Calgary, AB

- Analyzed building energy performance and thermal data for the MacKimmie Tower, delivering actionable optimization recommendations to improve operational efficiency. Supervised by Dr. David Wood.

Mechanical Design Intern May – August 2017
University of Alberta Edmonton, AB

- Designed and modelled an industrial multi-function CNC machine aimed to automate the fastening and cutting processes for wooden wall production. Supervised by Dr. Mohamed Al-Hussein.

Employment Experience

Teaching Assistant January – April 2021, 2022, 2023, 2025
University of Toronto Toronto, ON

- Delivered instruction and academic support to 400+ undergraduate and graduate students across 6 cohorts of mechanical engineering curriculum both online and in-person.

Project Engineering Intern August – December 2019
WSP in Canada Calgary, AB

- Developed a proprietary Class V capital cost estimation tool in Excel for industrial processing facilities, leveraging historical data and scalable process equipment models to improve budget forecasting accuracy.

Project Management Intern May – December 2018
City of Edmonton Edmonton, AB

- Planned and executed a major public engagement open house for a key infrastructure project, successfully presenting the design to 250+ community members and city councillors to secure critical stakeholder buy-in.

Equipment Reliability Engineering Intern September – December 2017
Syncrude Canada Ltd Fort McMurray, AB

- Engineered and delivered reliability packages to resolve both emergency equipment failures and scheduled maintenance, ensuring full compliance with ASME B31.3, B31.1, Sec. VIII Div. 1, and API 650 codes.

Talks, Presentations, and Other Involvement

Inflection Awardee March 10–11, 2025
Inflection Award Paris, France

- Selected as one of the 30 best young scientists in the world for breakthrough research in advancing climate technologies. Awardees were invited to a two-day program featuring high-profile speakers, workshops, and networking opportunities with top investors, entrepreneurs, and policymakers in climate tech.

Conference Presentation October 11, 2024
PRiME 2024 Honolulu, HI

- Reactive Capture of CO_2 via Amino Acid

Conference Presentation October 12, 2023
244th ECS Meeting Gothenburg, Sweden

- Integrated Capture and Electrochemical Conversion of CO_2 into CO

Awards and Scholarships

Inflection Award March 2025
2-day workshop Paris, France

Hatch Graduate Scholarship for Sustainable Energy Research April 2024
10,000 CAD University of Toronto

Natural Science & Engineering Research Council Post-graduate Scholarship May 2023
21,000 CAD, 40,000 CAD, and 40,000 CAD over 36 months University of Toronto

Hatch Graduate Scholarship for Sustainable Energy Research March 2023
10,000 CAD University of Toronto

Bert Wasmund Graduate Fellowships in Sustainable Energy Research December 2022
6,938 CAD University of Toronto

Hatch Graduate Scholarship for Sustainable Energy Research March 2022
10,000 CAD University of Toronto

Sustainable Development Goals Grant July 2020 – March 2021
5,000 CAD University of Calgary

Undergraduate Scholarship September 2019 – April 2020
1,500 CAD University of Alberta

Louise McKinney Post-Secondary Scholarship September 2019 – April 2020
2,500 CAD University of Alberta

Undergraduate Scholarship September 2017 – April 2018
2,500 CAD University of Alberta

Fluor Canada Ltd Award September 2017 – April 2018
2,500 CAD University of Alberta

T4K Undergraduate Scholarship September 2016 – April 2017
2,500 CAD University of Alberta

Jason Lang Scholarship September 2016 – April 2017
1,000 CAD University of Alberta

Enbridge Academic Excellence September 2015 – April 2016
2,125 CAD University of Alberta

Academic Excellence September 2015 – April 2016
4,875 CAD University of Alberta

ManCap Foundation Education September 2015 – April 2016
3,500 CAD University of Alberta

Peer-Reviewed Contributions

 # denotes equal contribution

1. **Xiao, Y.C.**, Sun, S.S., Miao, R.K., Han, K., Just, P., Corbett, P.J., Sinton, D. Industrializing reactive capture of CO_2 (under review)
2. **Xiao, Y.C.**, Guo, Z., Wang, R., Li, F., Sun, S.S., Liu, M., Lee, H.S., Liu, J., Gabardo, C.M., Wang, C., Villamanca, D.M., Yong, Z., Han, K., Miao, R.K., Corbett, P.J., Sinton, D. Solvent tuning regulates proton flux to extend stability in reactive CO2 capture and electrolysis. ***Chem. Catal.*** 2026, 6, 101694. DOI: 10.1016/j.checat.2026.101694
3. **Xiao, Y.C. #**, Sun, S.S. #, Zhao, Y., Miao, R.K., Fan, M., Lee, G., Chen, Y., Gabardo, C.M., Yu, Y., Qui, C., Guo, Z., Wang, X., Papangelakis, P., Huang, J.E., Li, F., O'Brien, C.P., Kim, J., Han, K., Corbett, P.J., Howe, J.Y., Sargent, E.H., Sinton, D. Reactive capture of CO_2 via amino acid. ***Nat. Comm.*** 2024, 15, 7849. DOI: 10.1038/s41467-024-51908-3
4. **Xiao, Y.C.**, Gabardo, C.M., Liu, S., Lee, G., Zhao, Y., O'Brien, C.P., Miao, R.K., Xu, Y., Edwards, J.P., Fan, M., Huang, J.E., Li, J., Papangelakis, P., Alkayyali, T., Rasouli, A.S., Zhang, J., Sargent, E.H., Sinton, D. Direct carbonate electrolysis into pure syngas. ***EES Catal.*** 2023, **1**, 54-61. DOI: 10.1039/d2ey00046f
5. Li, F.#, Guo, Z. #, Yan, Y. #, Wang, Q. #, Liang, Y., Duan, L., Bonnenfant, L., Dijvejin, Z.A., Liu, M., **Xiao, Y.C.**, Zhao, Z., Liu, J., Stepanovic, A., Gao, Y., Sun, Q., Abbas, H.G., Fan, M., Huang, J.E., Park, S., Holmes, S.M., Zhao, Y., Xu, Y., Dinh, C.-T., Wang, Z., Miao, R.K., Sinton, D. Electrosynthesis of ethylene from syngas. Nat. Sustain. 2026. DOI: 10.1038/s41893-025-01764-w
6. Sun, S.S. #, **Xiao, Y.C. #**, Li, F., Wu, J., Che, Y., Wang, Y., Liu, M., Guo, Y., Fan, M., Han, K., Just, P.-E., Corbett, P.J., Miao, R.K., Sinton, D. Industrial amine blends enable efficient CO electrolysis in reactive capture. ***EES Catal.*** 2025, 4, 387-396. DOI: 10.1039/d5ey00333d
7. Kim, J. #, Mahesh, S. #, Lee, H.S., Dorakhan, R., Bai, Y., Imran, M., Li, K., Liu, Y., Kim, D., Park, S., Zeraati, A.S., Moon, H.S., Li, X., Arabyarmohammadi, F., Abed, J., Wander, B., Wu, C., Liu, S., **Xiao, Y.C.**, Miao, R.K., Hoogland, S., Hattrick-Simpers, J., Sargent, E.H., Sinton, D. Accelerated discovery of CO_2-to-C_3-hydrocarbon electrocatalysts with human-in-the-loop. **Joule**. 2025, **9**, 102213. DOI: 10.1016/j.joule.2025.102213
8. Guo, Z. #, Li, F. #, **Xiao, Y.C. #**, Hung, S., Lu, Y., Foroozan, A., Liu, J., Sun, S.S., Liu, S., Che, Y., Wang, Q., Liu, M., Wang, C., Li, Y., Peng, K., Liu, Y., Fan, M., Dijvejin, Z.A., Papangelakis, P., Wang, Y., Zeraati, A.S., Han, K., Corbett, P.J., Higgins, D., Miao, R.K., Sinton, D. Efficient amino-acid-based reactive capture of CO_2 via nickel molecular catalyst. ***Nat. Comm.*** 2025, 16, 10373. DOI: 10.1038/s41467-025-65331-9
9. Miao, R.K. #, Fan, M. #, Wang, N. #, Zhao, Y., Li, F., Liu, M., Arabyarmohammadi, F., Liang, Y., Ni, W., Xie, K., Chen, Y., Sun, P., Huang, J.E., Wu, J., Kim, J., O'Brien, C.P., **Xiao, Y.C.**, Guo, Z., Papangelakis, P., Zeraati, A.S., Xu, Y., Dinh, C.-T., Sargent, E.H., Sinton, D. CO electrolysers with 51% energy efficiency towards C_{2+} products via the use of porous separators. ***Nat. Energy*** 2025, 10, 1197-1204. DOI: 10.1038/s41560-025-01846-1

10. Bodiuzzaman, M. #, Fan, L. #, Halappa, N.M. #, Chen, Y. #, Yan, Y. #, Thomas, S., Miao, R.K., **Xiao, Y.C.**, Wu, J., Dorakhan, R., Abulikemu, M., Tall, O.E., Sinton, D., Alshareef, H.N., Mohammed, O.F., Sargent, E.H., Bakr, O.M. High-Nuclearity Copper Molecular Catalysts for Electrocatalytic CO-to-Acetate Conversion. ***J. Am. Chem. Soc.*** 2025, 147, 31713. DOI: 10.1021/jacs.5c08144
11. Papangelakis, P. #, O'Brien, C.P. #, Zeraati, A.S. #, Liu, S., Paik, A., Nelson, V., Park, S., **Xiao, Y.C.**, Dorakhan, R., Sun, P., Wu, J., Gabardo, C.M., Wang, N., Miao, R.K., Sargent, E.H., Sinton, D. Scaled CO Electroreduction to Alcohols. ***Nat. Comm.*** 2025, **16**, 4969. DOI: 10.1038/s41467-025-59180-9
12. Zhang, J. #, Cao, Y. #, Ou, P. #, Lee, G. #, Zhao, Y. #, Liu, S., Shirzadi, E., Dorakhan, R., Xie, K., Tian, C., Chen, Y., Li, X., **Xiao, Y.C.**, Zeraati, A.S. Miao, R.K., Park, S., O'Brien, C.P., Ge, J., Zhou, X., Sinton, D., Sargent, E.H. A redox-active polymeric network facilitates electrified reactive-capture electrosynthesis to multi-carbon products from dilute CO_2-containing streams. ***Nat. Comm.*** 2025, **16**, 3553. DOI: 10.1038/s41467-025-58756-9
13. Guo, Y., Li, F., Saber, S., Zargartalebi, M., Sun, S.S., **Xiao, Y.C.**, Bao, B., Xu, Z., Sinton, D. Rapid screening of CO_2 capture fluids. ***Lab Chip*** 2025, 25, 2918. DOI: 10.1039/d4lc00772g
14. O'Brien, C.P. #, McLaughlin, D. #, Bohm, T., **Xiao, Y.C.**, Edwards, J.P., Gabardo, C.M., Bierling, M., Wicks, J., Rasouli, A.S., Abed, J., Young, D., Dinh, C., Sargent, E.H., Thiele, S., Sinton, D. Scalability and stability in CO_2 reduction via tomography-guided system design. ***Joule*** 2024, **8**, 2903-2919. DOI: 10.1016/j.joule.2024.07.004
15. Papangelakis, P. #, Miao, R.K. #, Lu, R. #, Liu, H. #, Wang, X. #, Ozden, A., Liu, S., Sun, N., O'Brien, C.P., Hu, Y., Shakouri, M., Xiao, Q., Li, M., Khatir, B., Huang, J.E., Wang, Y., **Xiao, Y.C.**, Li, F., Zeraati, A.S., Zhang, Q., Liu, P., Golovin, K., Howe, J.Y., Liang, H., Wang, Z., Li, J., Sargent, E.H., Sinton, D. Improving the SO_2 tolerance of CO_2 reduction electrocatalysts using a polymer/catalyst/ionomer heterojunction design. ***Nat. Energy*** 2024, **9**, 1011-1020. DOI: 10.1038/s41560-024-01577-9
16. Liu, S. #, Zhang, J. #, Li, F. #, Edwards, J.P. #, **Xiao, Y.C.**, Kim, D., Papangelakis, P., Kim, J., Elder, D., De Luna, P., Fan, M., Lee, G., Miao, R.K., Yan, Y., Chen, Y., Zhao, Y., Guo, Z., Tian, C., Li, P., Xu, Y., Sargent, E.H., Sinton, D. Direct Air Capture of CO_2 via Cyclic Viologen Electrocatalysis. ***Energy Environ. Sci.*** 2024, **17**, 1266-1278. DOI: 10.1039/D3EE03024E
17. Alerte, T. #, Goana, A. #, Edwards, J.P. #, Gabardo, C.M., O'Brien, C.P., Wicks, J., Bonnenfant, L., Rasouli, A.S., Young, D., Abed, J., Kershaw, L., **Xiao, Y.C.**, Sarkar, A., Jaffer, S.A., Schreiber, M.W., Sinton, D., MacLean, H.L., Sargent, E.H. Scale-Dependent Techno-Economic Analysis of CO_2 Capture and Electroreduction to Ethylene. ***ACS Sustainable Chem. Eng.*** 2023, **11**, 15651-15662. DOI: 10.1021/acssuschemeng.3c04373
18. Miao, R.K. #, Wang., N. #, Hung, S., Huang, W., Zhang, J., Zhao, Y., Ou, P., Wang, S., Edwards, J.P., Tian, C., Han, J., Xu, Y., Fan, M., Huang, J.E., **Xiao, Y.C.**, Ip, A.H., Liang, H., Sargent, E.H., Sinton, D. Electrified Cement Production via Anion-Mediated Electrochemical Calcium Extraction. ***ACS Energy Lett.*** 2023, **8**, 4694-4701. DOI: 10.1021/acsenergylett.3c01668
19. Xu, Y. #, Liu, S. #, Edwards, J.P., **Xiao, Y.C.**, Zhao, Y., Miao, R.K., Fan, M., Chen, Y., Huang, J.E., Sargent, E.H., Sinton, D. Regeneration of direct air CO_2 capture liquid via alternating electrocatalysis. ***Joule*** 2023, **7**, 2107-2117. DOI: 10.1016/j.joule.2023.07.011
20. Fan, M. #, Huang, J.E. #, Miao, R.K. #, Mao, Y. #, Ou, P.#, Li, F., Li, X., Cao, Y., Zhang, Zishuai, Zhang, J., Yan, Y., Ozden, A., Ni, W., Wang, Y., Zhao, Y., Chen, Z., Khatir, B., O'Brien, C.P., Xu, Y., **Xiao, Y.C.**, Waterhouse, G.I, Golovin, K., Wang, Z., Sargent, E.H., Sinton, D. Cationic-group-functionalized electrocatalysts enable stable acidic CO_2 electrolysis. ***Nat. Catal.*** 2023, **6**, 763-772. DOI: 10.1038/s41929-023-01003-5
21. Lee. G. #, Rasouli, A.S. #, Lee, B. #, Zhang, J., **Xiao, Y.C.**, Edwards, J.P., Lee, M.G., Jung, E.D., Arabyarmohammadi, F., Liu H., Grigioni, I., Abed, J., Alkayyali, T., Liu, S., Xie, K., Miao, R.K., Park, S., Dorakhan, R., Zhao, Y., O'Brien, C.P., Chen, Z., Sinton, D., Sargent, E.H. CO_2 electroreduction to multicarbon products from carbonate capture liquid. ***Joule*** 2023, **7**, 1277-1288. DOI: 10.1016/j.joule.2023.05.003
22. Fan, M. #, Miao, R.K. #, Ou, P. #, Xu, Y. #, Lin, Z., Lee, T., Hung, S., Xie, K., Huang, J.E., Ni, W., Li, J., Zhao, Y., Ozden, A., O'Brien, C.P., Chen, Y., **Xiao, Y.C.**, Liu, S., Wicks, J., Wang, X., Abed, J., Shizadi, E., Sargent, E.H., Sinton, D. Single-site decorated copper enables energy- and carbon-efficient CO_2 methanation in acidic conditions. ***Nat. Commun.*** 2023, **14**, 3314. DOI: 10.1038/s41467-023-38935-2
23. Edwards, J.P., Alerte, T., O'Brien, C.P., Gabardo, C.M., Liu, S., Wicks, J., Gaona, A., Abed, J., **Xiao, Y.C.**, Young, D., Rasouli, A.S., Sarkar, A., Jaffer, S.A., MacLean, H.L., Sargent, E.H., Sinton, D. Pilot-Scale CO_2 Electrolysis Enables a Semi-empirical Electrolyzer Model. ***ACS Energy Lett.*** 2023, **8**, 2576-2584. DOI: 10.1021/acsenergylett.3c00620

24. Kose, T., O'Brien, C.P., Wicks, J., Abed, J., **Xiao, Y.C.**, Sutherland, B., Sarkar, A., Jaffer, S.A., Sargent, E.H., Sinton, D. High-throughput parallelized testing of membrane electrode assemblies for CO_2 reduction. ***Catal. Sci. & Technol.*** 2022, **12**, 6239-6245. DOI: 10.1039/d2cy00873d